DNA Structure: Pair, Twist, Critique

LAUNCH · Science · 40 minutes

I can construct a complementary DNA section and describe the strengths and limitations of the model.

Prepare before pupils arrive

Setup: Prepare large base cards and a pre-checked Standard sequence. Keep the base-pair key visible.

Safety: Check component size/material access and provide a flat model route for pupils who do not want twisting or handling.

Equipment: Large nucleotide/base cards, reusable backbone strips, clips or hook-and-loop tabs, sequence sheet and pencil.

Safety: Use large components and avoid latex or small clips where local needs require. Do not wrap the model around a person.

Fallback: Direct an adult to place each base or point to the partner; dictate the sequence and model limitation.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe DNA as a polymer of nucleotides arranged as two strands coiled into a double helix, with complementary base pairs joined by weak hydrogen bonds.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Describe the chromosome-DNA-gene relationship. Define genome.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	nucleotide — one DNA building unit complementary — fits by a pairing rule double helix — two-strand twist DNA is a polymer of nucleotides. Each nucleotide contains a sugar, a phosphate group and one base; nucleotides join to form each sugar-phosphate backbone.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Strength: the ladder shows two strands, base order and complementary pairing clearly. Limit: colours, distances and component sizes are invented; a classroom twist is not the molecule's true scale or movement.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 5 min	Match each base to its complementary partner, freeze the sequence, then critique one feature of the model. READ PAIR CHECK TWIST CRITIQUE Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 12 min	Construct a complementary DNA section, annotate its structure and evaluate the model. Record: A correct complementary sequence, labelled two-strand/double-helix description and one valid model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Write the complement of A-G-C-T. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: C-T-A-G

Yes. G pairs with C, A with T, T with A and C with G.

Misconception repair

Repaired. The strands are complementary because each base pairs by a fixed rule, not because the letters are identical.

Guided checkpoint

Yes. G pairs with C, A with T, T with A and C with G. Apply one base-pair rule at a time from left to right.

Adenine

A ↔ T PAIR. A-T pair completed. Use the fixed pairing rule for adenine.

Cytosine

C ↔ G PAIR. C-G pair completed. Use the fixed pairing rule for cytosine.

Not to scale

MODEL LIMIT. Model limitation identified. Ask whether the classroom size matches molecular scale.

Model limitation

The model is enormously enlarged, uses arbitrary colours, simplifies chemical bonds and cannot show the full chromosome or molecular dynamics.

Independent evidence

A correct complementary sequence, labelled two-strand/double-helix description and one valid model limitation.

Supported: Complete a six-base sequence with A-T and C-G cards, then point to the two strands. Pairing key stays visible; one base at a time and exact scribing are allowed.

Standard: Complete a ten-base complementary sequence and describe nucleotide, two strands and double helix. Use: Base __ pairs with __; the two strands __.

Stretch: Construct and check the sequence, then evaluate two useful features and two limitations of the model. Do not add Higher-tier protein synthesis; deepen model evaluation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.

What the learner actually showed	Next teaching action
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W12L2_DNA_Structure_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.